Multi-microphone and multi-modal enhancement of speech conversations for assistive listening

Position: Master 2 / end-of-studies internship.
Duration: 5 to 6 months, starting in February or March 2027.
Location: CentraleSupélec (campus of Rennes), Cesson-Sévigné, France.
Affiliation: AIMAC team of the IETR laboratory (UMR CNRS 6164).
Supervisors: Louis Bahrman, Sofiene Kammoun, Simon Leglaive

Context

The human auditory system shows outstanding performance at focusing on speech in the presence of background noise. In social contexts, the so-called “Cocktail party effect” describes the ability of the brain to focus on a single speech source in the presence of competing speakers. Yet, people suffering from hearing impairment struggle to understand their pairs in noisy acoustic environments.

Speech enhancement is the task that aims to extract a clean speech signal from a noisy acoustic recording. It represents a vast research field of audio signal processing. Cocktail party scenarios have been the object of numerous studies and challenges for the audio processing community. While most speech enhancement works focus on enhancing a single speaker, recent works have considered enhancing multiple speakers within the same conversation (Leglaive et al., 2023). In particular, the *Enhancing Conversation for Hearing Inclusivity* (ECHI) task of the CHiME-9 challenge has been recently proposed to advance and benchmark speech enhancement methods in such context, by enhancing a full conversation in the presence of background noise (Sutherland et al., 2026).

This task is motivated by an assistive listening use case, in which a speech enhancement algorithm can help any individual to better engage in a conversation, by improving the overall conversation intelligibility and quality within the ambient noise.

Objectives

The overall objective of this internship is to propose a method to address the ECHI-2 task of the CHiME-10 challenge, which focuses on enhancing conversations recorded with wearable smart glasses to improve listening in noisy social environments. This task provides a multimodal dataset of real conversations, combining multichannel audio with synchronized video and motion sensors, and includes both audio-only and multimodal challenge tracks. The developed system should be low-latency and it should produce an enhanced binaural rendering of the conversation by improving the intelligibility and quality of conversational speech while suppressing background noise, competing talkers, and the wearer’s own voice.

Several tasks will be conducted during this internship:

- State-of-the-art methods for multi-channel speech enhancement or extraction and previous ECHI challenge submissions will be reviewed. This literature review will include neural approaches such as TF-GridNet (Wang et al., 2023) or USES (Zhang et al., 2023) as well as signal model-based methods (Duong et al., 2010) and combinations of both (Munakata et al., 2023).
- Based on this literature review and the official baseline of the challenge, a system will be developed to address the ECHI-2 task. The system should be low latency, and it should exploit spatial information provided by the multi-microphone recording (Tesch and Gerkmann, 2024).
- If aiming for the multimodal track of the challenge, video and motion sensors could be used in combination with the multichannel audio signals (Michelsanti et al., 2021).
- The proposed method will be experimentally validated using objective metrics on the ECHI-2 challenge dataset, and potentially be submitted as an entry to the challenge.

Candidate profile

The candidate will be pursuing their last year of Master's or engineer's degree. They should have good knowledge and practical skills in machine / deep learning and signal processing. Proficiency in Python is required. Experience with PyTorch as well as past projects in speech or audio processing would be a plus. The candidate should also have good oral and written communication skills, and be self-motivated.

How to apply

Interested candidates should apply by filling out this form before November 22, 2026:

Candidates will be invited to upload (as PDF files) a resume, the official transcripts for each year of higher education, and a short motivation letter providing evidence of the candidate's interest for the topic of the internship (e.g., past student project in audio or speech processing).

In case of difficulty or question, please contact louis.bahrman@centralesupelec.fr.

Work environment

The intern will be supervised by Louis Bahrman, Sofiene Kammoun and Simon Leglaive, and will integrate the AIMAC team of the IETR laboratory, located in CentraleSupélec's campus of Rennes (in Brittany, France).

CentraleSupélec offers accommodation on the campus.

The intern will benefit from the research environment of CentraleSupélec, in particular the computational resources of the Mésocentre.

The intern will receive the legal internship gratification of about 630 €/month.

References

Leglaive, S., et al. (2023). "The CHiME-7 UDASE task: Unsupervised domain adaptation for conversational speech enhancement". International Workshop on Speech Processing in Everyday Environments (CHiME).

Sutherland, Robert, et al. (2026). "CHiME-9 ECHI: A Machine Learning Challenge for Enhancing Conversations to Address Hearing Impairment." International Workshop on Acoustic Signal Enhancement (IWAENC).

Wang, Z. Q., Cornell, S., Choi, S., Lee, Y., Kim, B. Y., & Watanabe, S. (2023). TF-GridNet: Integrating full- and sub-band modeling for speech separation. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 31, 3221-3236.

Zhang, W., Saijo, K., Wang, Z. Q., Watanabe, S., & Qian, Y. (2023). Toward universal speech enhancement for diverse input conditions. In IEEE Automatic Speech Recognition and Understanding Workshop (ASRU) (pp. 1-6).

Duong, N. Q., Vincent, E., & Gribonval, R. (2010). Under-determined reverberant audio source separation using a full-rank spatial covariance model. IEEE Transactions on Audio, Speech, and Language Processing, 18(7), 1830-1840.

Munakata, H., Bando, Y., Takeda, R., Komatani, K., & Onishi, M. (2023). Joint separation and localization of moving sound sources based on neural full-rank spatial covariance analysis. IEEE Signal Processing Letters, 30, 384-388.

Tesch, K., & Gerkmann, T. (2023). Multi-channel speech separation using spatially selective deep non-linear filters. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 32, 542-553.

Michelsanti, D., Tan, Z. H., Zhang, S. X., Xu, Y., Yu, M., Yu, D., & Jensen, J. (2021). An overview of deep-learning-based audio-visual speech enhancement and separation. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 29, 1368-1396.